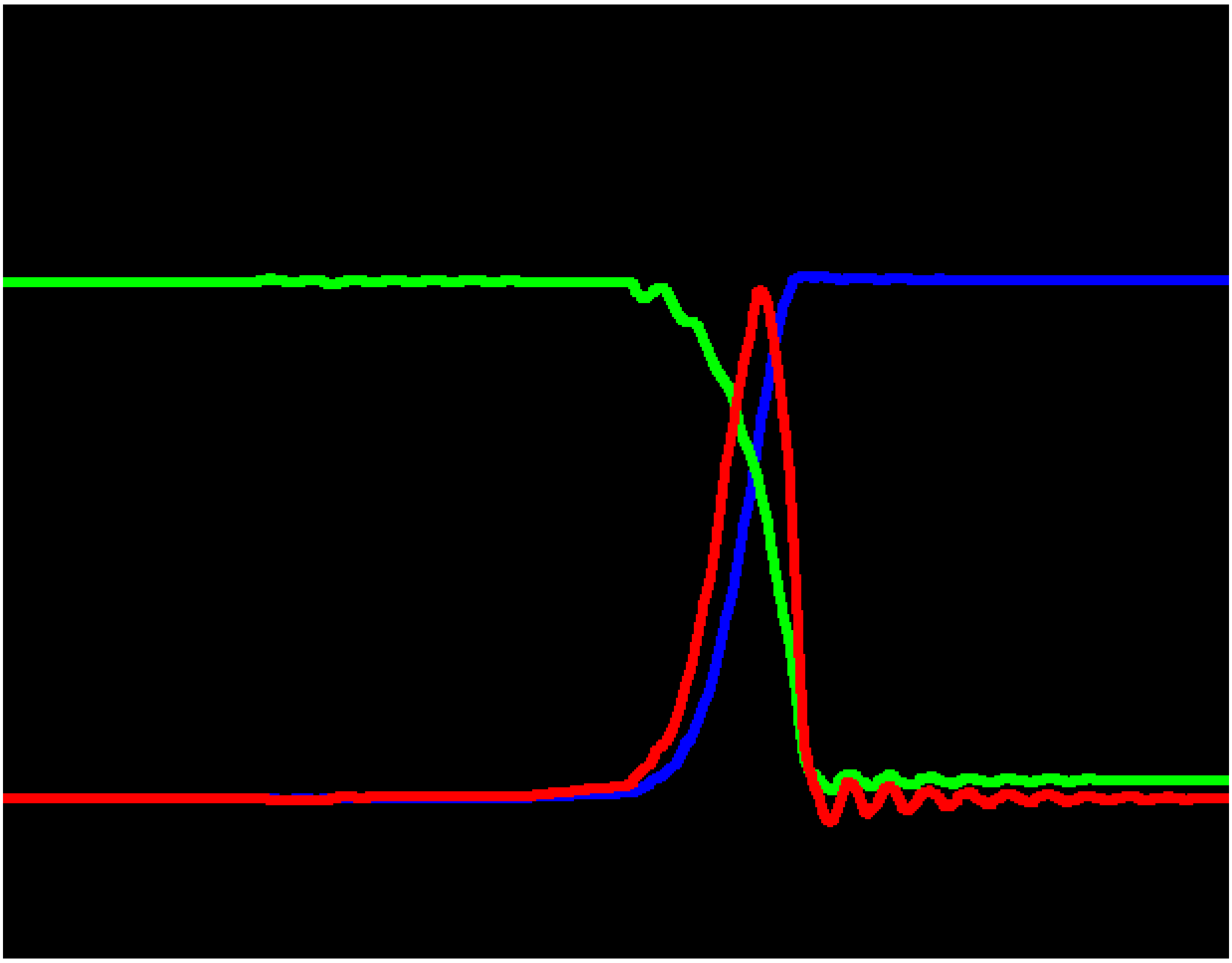
Mosfets And Drivers



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Primary

Mosfet

Introduction

In the first version of the BOM, the primary mosfet was **IPW60R037CM8XKSA1 600 V 64A 37mΩ**
The second and final version of the primary mosfet is **IPW60R037P7 600V 76A 37mΩ**

for mor information, see the Mosfet IPW60R037P7 datasheet.

From the initial calculation of the LLC thank , we have the the input data for primary mosfet calculation:

```
Lnc                3.0
Qec                0.55
Cr_nF              116.209
n                  4.0
Lr_uH              21.797
Lm_uH              65.392
fsw_min            60170.0
fsw_max            156220.0
Im_rms             6.992
Io                 25.0
Ioe_rms            7.636
Ios_rms            30.545
Ir_rms             10.354
L_second_uH        4.087
Re_nom             24.901
Re_110             22.637
Cr                 1.16e-07
Lr                 2.18e-05
Lm                 6.54e-05
```

We have the rms current in primary side:

$$\mathrm{Ir}_{rms} = 10.354 \ \ (\mathrm{Arms})$$

So the rms current in the moset is

$$I_{Q\_rms} = \sqrt{\frac{1}{2} I_{r\_rms}^2} = \frac{I_{r\_rms}}{\sqrt{2}}$$

Calculation

$$I_{Qrms} = \frac{\mathrm{Ir}_{rms}}{\sqrt{2}} = \frac{10.354}{\sqrt{2}} \qquad = 7.321 \ \ (\mathrm{Arms})$$

$$I_{Qpeak} = I_{Qrms} \cdot \sqrt{2} = 7.321 \cdot \sqrt{2} \qquad = 10.354 \ \ (\mathrm{Arms})$$

Parasitic capacitance of the mosfet

Cds:

In this part, we will remind the Coss calculation.

- Why: To understand the difference between the energy/time related capa, and calculate it if it not given using ONLY the curve Coss = f(Vds).

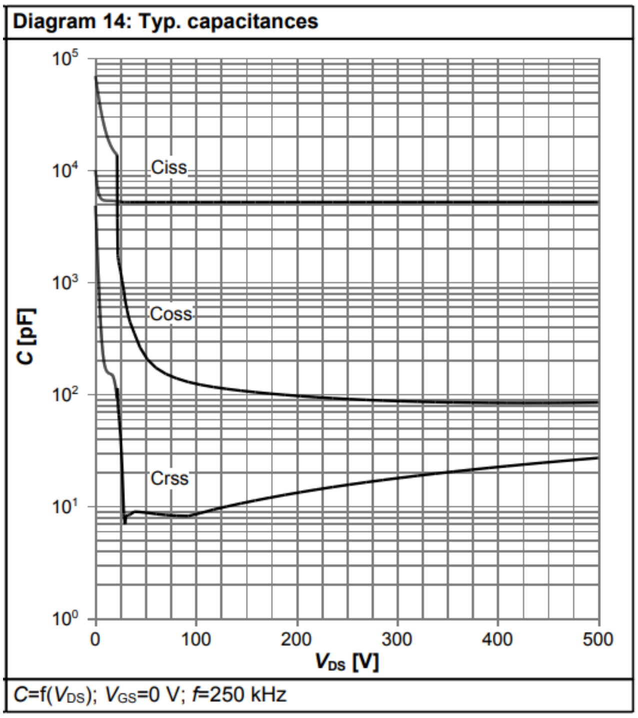


Figure 1 Datasheet capacitors curve as function of Vds

We use the tool that we are developed to extract the data from the image:

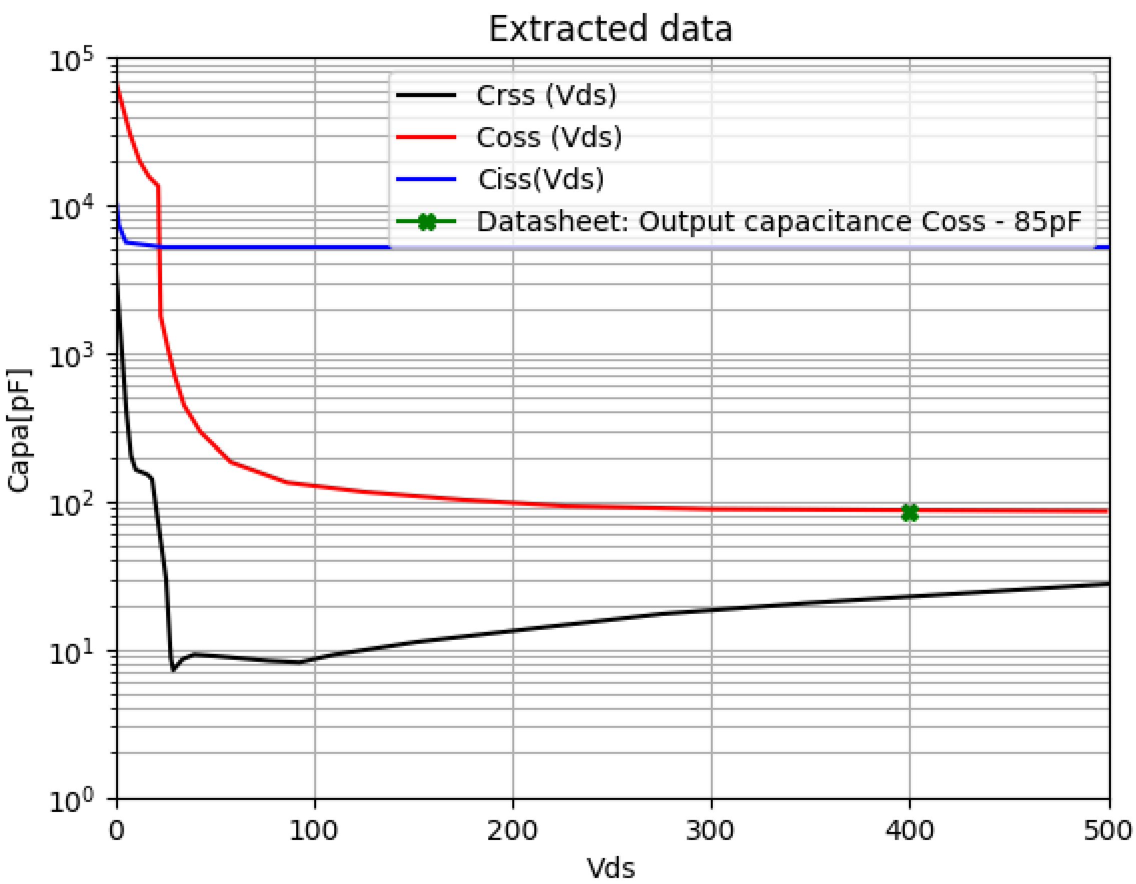


Table 5 Dynamic characteristics

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Input capacitance	C_{iss}	-	5243	-	pF	V_{GS}=0V, V_{DS}=400V, f=250kHz
Output capacitance	C_{oss}	-	85	-	pF	V_{GS}=0V, V_{DS}=400V, f=250kHz
Effective output capacitance, energy related[1)]	$C_{o(er)}$	-	156	-	pF	V_{GS}=0V, V_{DS}=0...400V
Effective output capacitance, time related[2)]	$C_{o(tr)}$	-	1599	-	pF	I_D=constant, V_{GS}=0V, V_{DS}=0...400V

Figure 2 Datasheet avg capacitors

Reminder:

Charge: $Q = C \cdot V \implies dQ = C(v) \cdot dv$

Energy: $E = \frac{1}{2} C \cdot V^2 \implies dE = v \cdot dQ = C(v) \cdot v \cdot dv$

Time-related ($C_{o(tr)}$):
The constant capacitance that gives the same charging time as the non-linear C_{oss} when charged with a constant current.

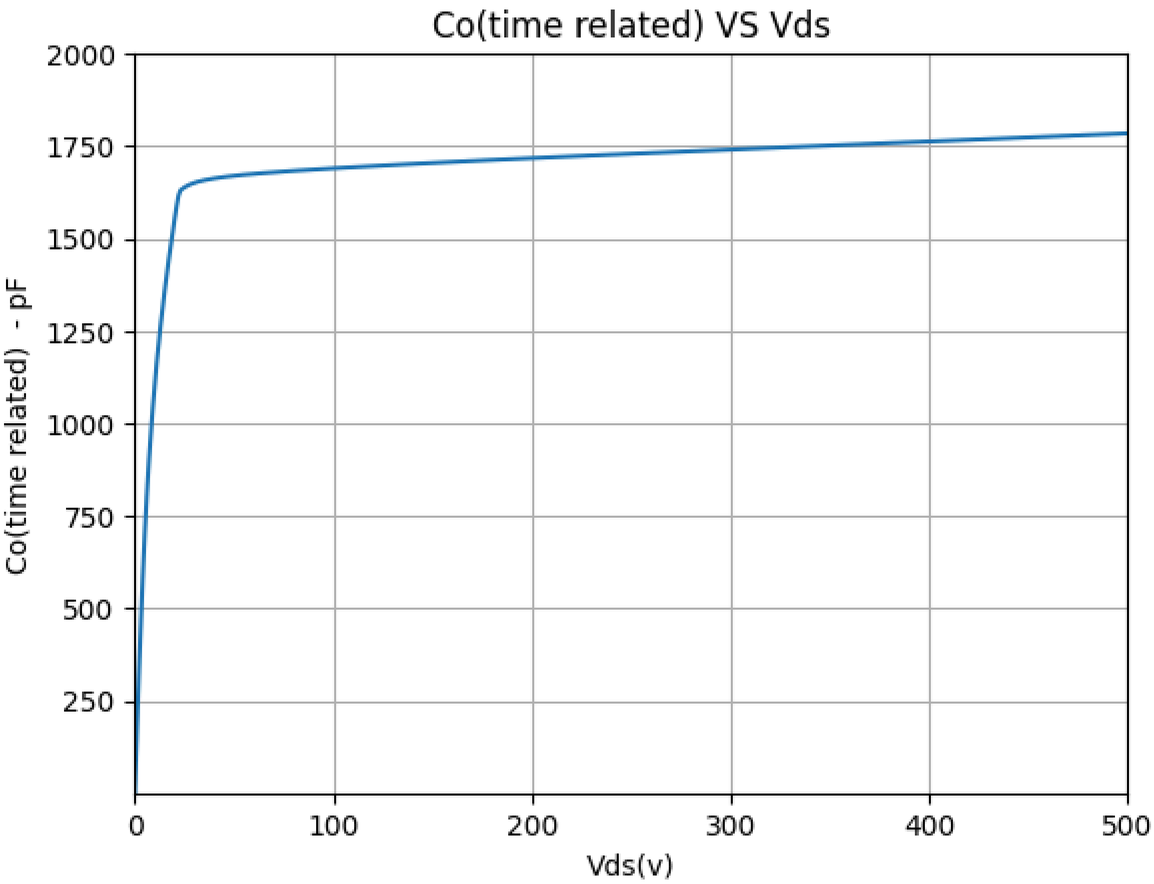
$$C_{o(tr)} = \frac{Q}{V_{DS}} = \frac{1}{V_{DS}} \int_0^{V_{DS}} C_{oss}(v)\, dv$$

Energy-related ($C_{o(er)}$):
The constant capacitance that gives the same stored energy as the non-linear C_{oss} at a destination voltage (V_{DS}).

$$C_{o(er)} = \frac{2E}{V_{DS}^2} = \frac{2}{V_{DS}^2} \int_0^{V_{DS}} C_{oss}(v) \cdot v\, dv$$

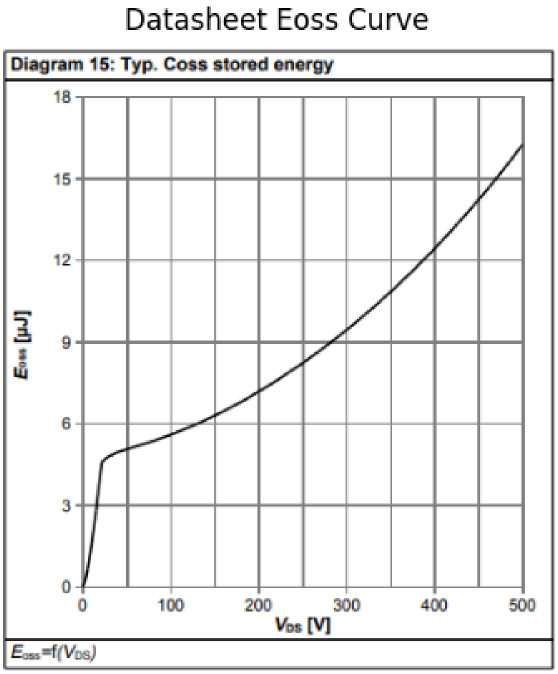
```
Total Qoss : 675.93 nC
Total Eoss : 12.77 µJ
Co(tr)     : 1689.82 pF  (Datasheet: 1599 pF)
Co(er)     : 159.64 pF  (Datasheet: 156 pF)
```

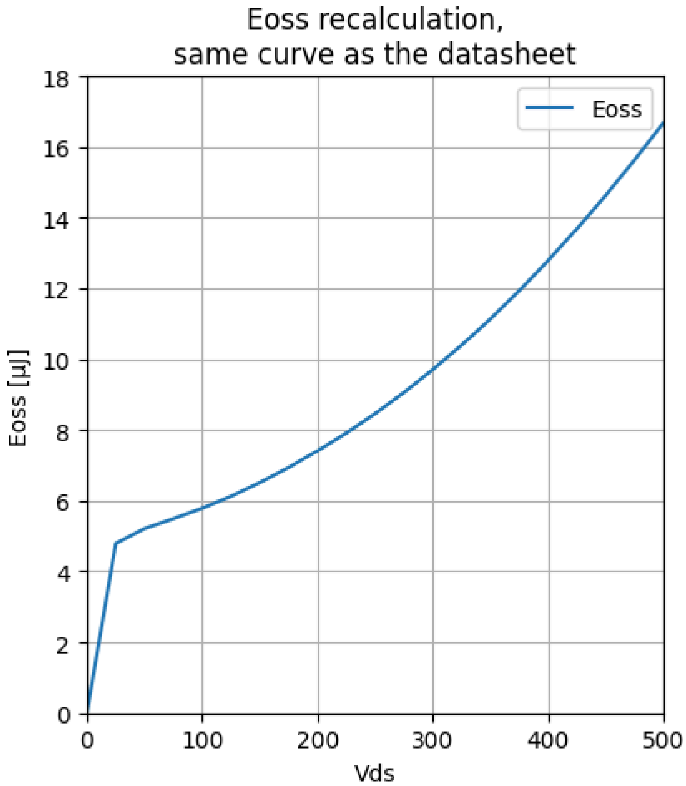
Let's make the same calculation of all voltage values (Vds)



We can use the below formula to calculate the energy stored in the C_{oss}:

$E_{oss} = \int_0^{V_{DS}} C_{oss}(v) \cdot v\, dv$





Gate charges

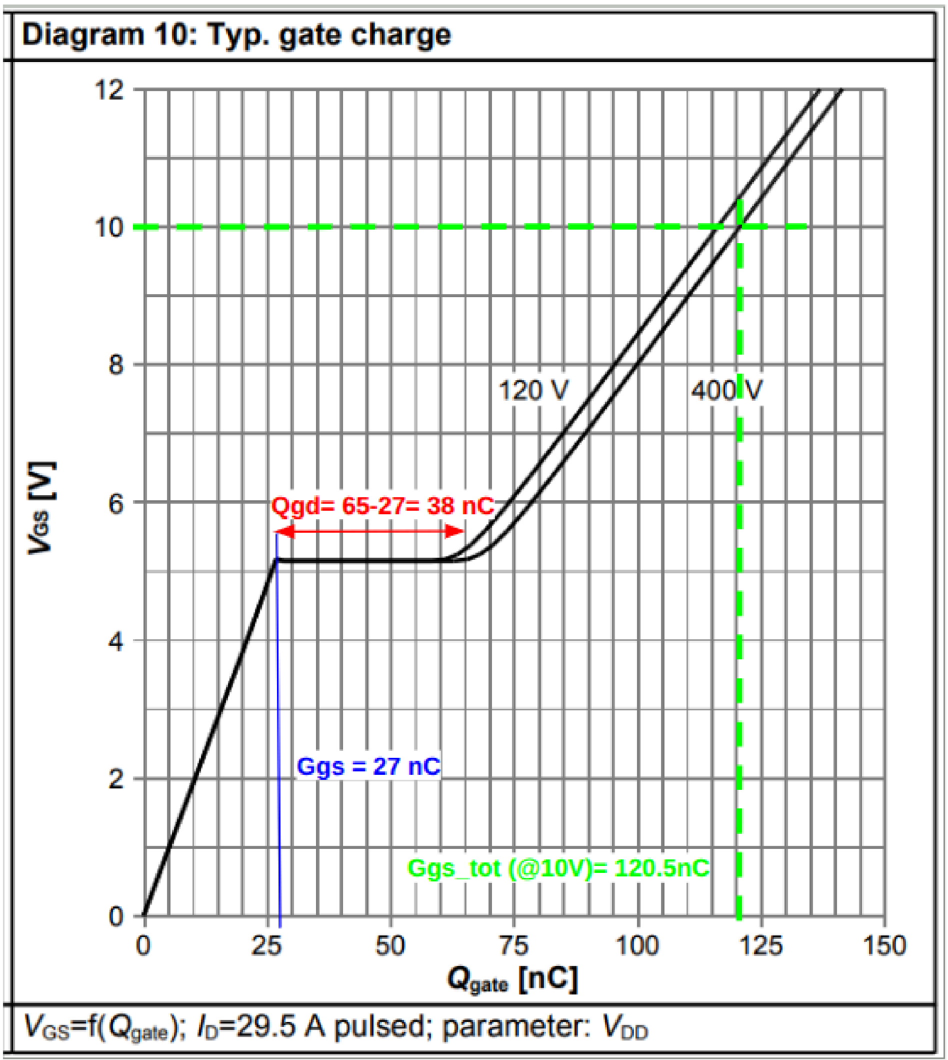


Figure 26 Datasheet: Plateau + The estimated gate charges

We can found almost the same value of the below table

Table 6 Gate charge characteristics

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Gate to source charge	Q_{gs}	-	27	-	nC	V_{DD}=400V, I_D=29.5A, V_{GS}=0 to 10V
Gate to drain charge	Q_{gd}	-	37	-	nC	V_{DD}=400V, I_D=29.5A, V_{GS}=0 to 10V
Gate charge total	Q_g	-	121	-	nC	V_{DD}=400V, I_D=29.5A, V_{GS}=0 to 10V
Gate plateau voltage	$V_{plateau}$	-	5.2	-	V	V_{DD}=400V, I_D=29.5A, V_{GS}=0 to 10V

Figure 5 Datasheet gate charges

Dead time

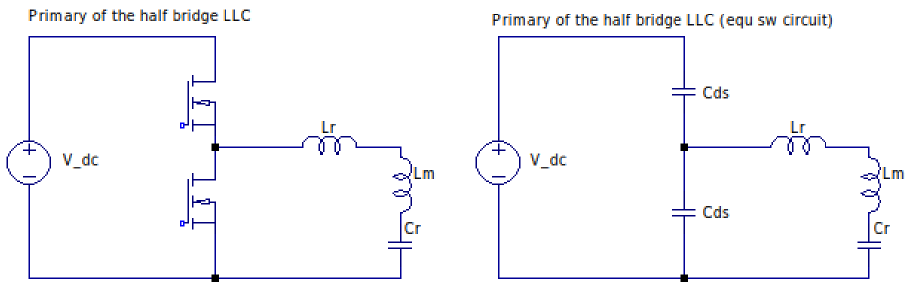


Figure 27 Switching equivalent circuit

The idea of dead time condition to assurate the ZVS is to discharge the parasite capacitor of the mosfet completely during the dead time.

The fundamental formula of the capacitor

$$i = C\frac{dV}{dt}$$

In mosfet leg ($C = 2.C_{DS}$: we must discharge the heigh side, and charge the low side, so 2 x)

$$Time_{discharge} = 2.C_{DS}\frac{V_{DC}-0}{I_{max}} = 2.C_{DS}\frac{V_{DC}}{I_{max}}$$

dead time must be :

$$Dead_{Time} > Time_{discharge}$$

The magnetic inductance define the maximum current

$$V_L = Lm\frac{di}{dt}$$

In half of periode current go from $-I_{max}$ to I_{max} in half of periode (Triangle waveform)

$$V_{DC} = Lm\frac{2.I_{max}}{T/2} = 4.Lm.\,I_{max}f_{sw}$$

This calculation take in account the 0 load hypothesis, the load increase I_{max} , so no load is the worst case for the ZVS condition.

Let's replace the I_{max}

$$Dead_{Time} > C_{DS}\frac{V_{DC}.4.Lm.f_{sw}}{V_{DC}}$$

And (half bridge)

$$\mathbf{Dead_{Time} > 8.Lm.\,f_{sw}.\,C_{DS}}$$

For full bridge voltage change from $-V_{DC}$ to V_{DC}, so $dV = 2.V_{DC}$ and the formula become

$$\mathbf{Dead_{Time} > 16.Lm.\,f_{sw}.\,C_{DS}}$$

For more details about the below input data, please check the LLC tank definition document.

$$Lm_{uH} = 65.392 \ \ (\mu H)$$
$$fsw_{min} = 60 \ \ (kHz)$$
$$Co_{tr} = 1599 \ \ (pF, IPW60R037P7, datasheet)$$
$$Dead_{TimeMin} = 8 \cdot Lm_{uH} \cdot fsw_{min} \cdot Co_{tr} \cdot 1 \times 10^{-6} = 8 \cdot 65.392 \cdot 60 \cdot 1599 \cdot 1 \times 10^{-6} = 50.190 \ \ (ns)$$

We will start with **100 ns** as dead time

Power losse related to the dead time

During dead time, the conduction is accurate with the body diode, so in the worst case (we suppose that the switching is instantly and during the dead time, the current pass in the body diode)

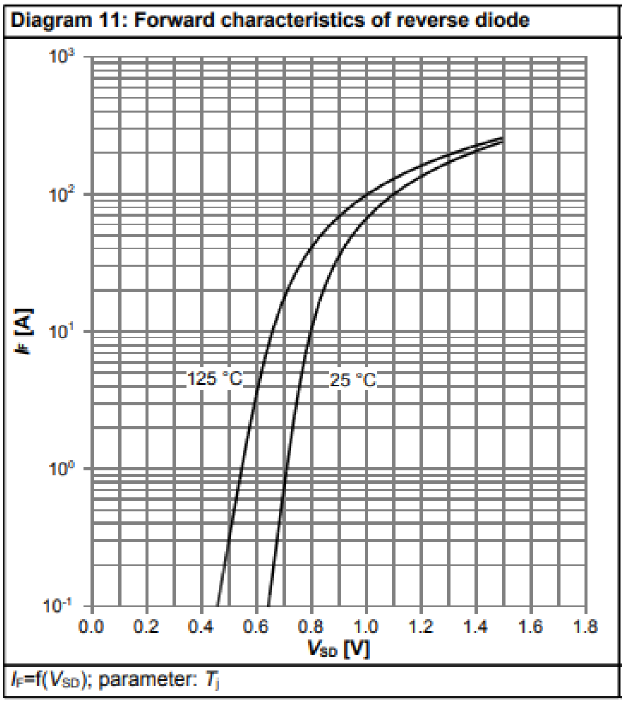


Figure 7 IPW60R037P7 datasheet: Vf

At 10-11A (primary current during the dead time), the Vf = 0.8V (worst case @ Tj = 25°C)

So the dissipated energy during the dead time is

$$dead_{time} = 0.100 \ \ (\mu s)$$
$$Vf = 0.800$$
$$Id = 11 \ \ (A)$$
$$E_{dead} = Vf \cdot Id \cdot dead_{time} = 0.800 \cdot 11 \cdot 0.100 \qquad = 0.880 \ \ (\mu J)$$

Since we have 2 dead time in each periode

$f_{swMax} = 150 \ \ \text{(kHz)}$

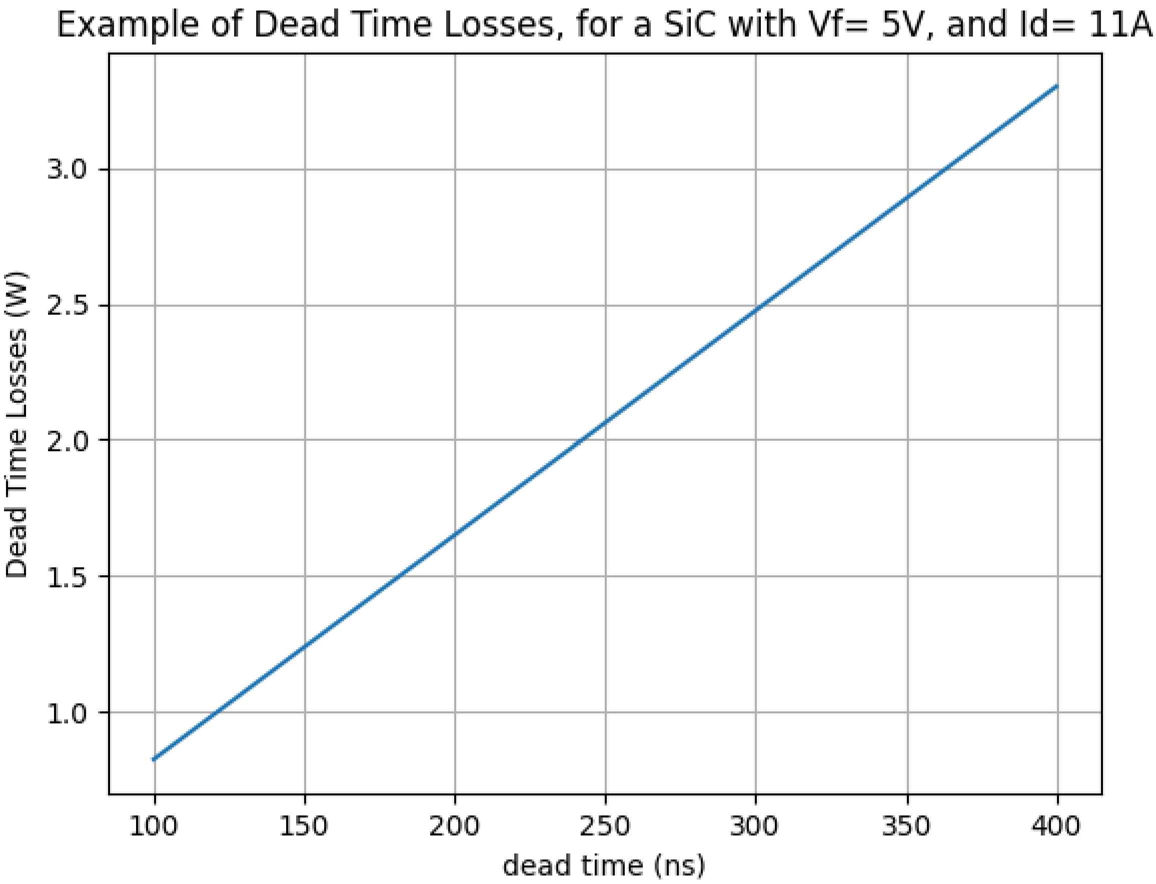
$P_{dead} = 1 \times 10^{-3} \cdot E_{dead} \cdot f_{swMax} = 1 \times 10^{-3} \cdot 0.880 \cdot 150 \qquad = 0.132 \ \ \text{(w)}$

It is relatively low, but if we use a SiC mosfet with heigh diode forword voltage like 5V, the power of the dead time will be

$\text{Vf} = 5$

$E_{dead} = \text{Vf} \cdot \text{Id} \cdot \text{dead}_{time} = 5 \cdot 11 \cdot 0.100 \qquad = 5.500 \ \ (\mu\text{J})$

$P_{dead} = 1 \times 10^{-3} \cdot E_{dead} \cdot f_{swMax} = 1 \times 10^{-3} \cdot 5.500 \cdot 150 \qquad = 0.825 \ \ \text{(w)}$



Because of this, the dead time must be keeped as lower as possible

Mosfet loss (Switching)

Since we will design the LLC to switch with ZWS so:

- Eon : Almost 0J (ZVS)
- Eoff : for low current, we will take the first approximation as Eoss

$$\mathrm{Vdc} = 400$$
$$\mathrm{Co}_{en} = 156 \ \ \mathrm{(pF)}$$
$$F_{sw} = 150 \ \ \mathrm{(kHz - Maximum)}$$
$$E_{off} = \frac{1}{2} \cdot 1 \times 10^{-6} \cdot \mathrm{Co}_{en} \cdot (\mathrm{Vdc})^2 = \frac{1}{2} \cdot 1 \times 10^{-6} \cdot 156 \cdot (400)^2 \qquad = 12.480 \ \ (\mu\mathrm{J})$$
$$P_{off} = 1 \times 10^{-3} \cdot F_{sw} \cdot E_{off} = 1 \times 10^{-3} \cdot 150 \cdot 12.480 \qquad = 1.872 \ \ (\mathrm{W})$$

We can use the datasheet Eoss curve to found the same value of Eoss = 12.48µJ (@400V)

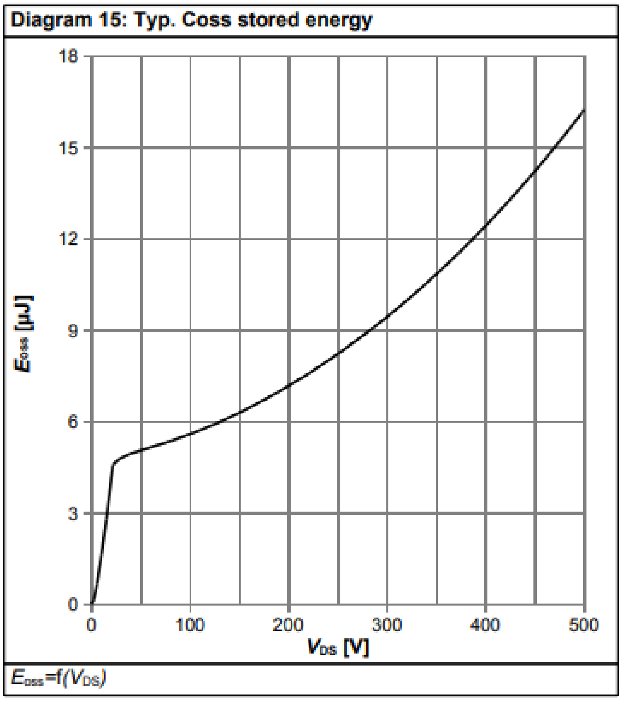


Figure 8 Datasheet avg capacitors

Since half bridge, the voltage is Vdc/2=200V

- Above curve ==> $Eoss = 7.5\mu J$

$$E_{off} = 7.500 \ \ (\mu\mathrm{J})$$
$$P_{off} = 1 \times 10^{-3} \cdot F_{sw} \cdot E_{off} = 1 \times 10^{-3} \cdot 150 \cdot 7.500 \qquad = 1.125 \ \ (\mathrm{W})$$

https://www.vishay.com/docs/73217/an608a.pdf

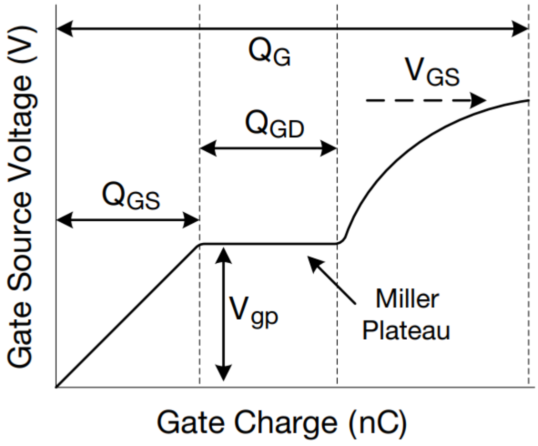


Figure 9 Sketch Showing Breakdown of Gate Charge

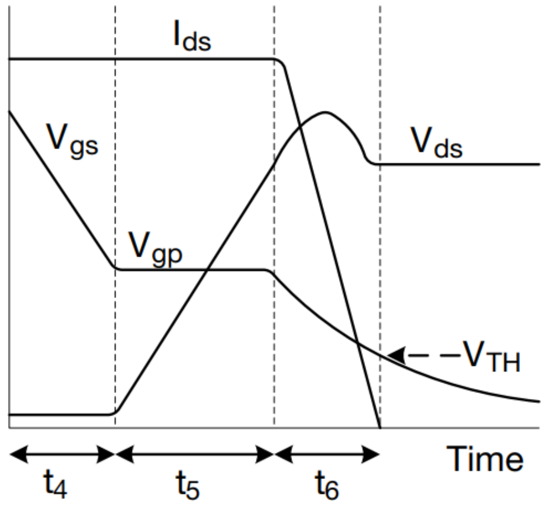


Figure 10 Turn-Off Transient of the MOSFET

To calculate the times of trun off, we will use the below formulas

For more information see Vishay document

Similarly for the turn-off transition, the voltage rise time ($t_{vr} = t_5$) is:

$$t_{vr} = t_5 = R_G \times \frac{Q_{GD(D)}}{V_{DS(D)}} \times \frac{V_{DS}}{V_{gp}} \qquad (19)$$

The current fall time ($t_{if} = t_6$) is:

$$t_{ir} = t_6 = R_G C_{iss\ at\ V_{DS}} \times \ln\left(\frac{V_{gp}}{V_{TH}}\right) \qquad (20)$$

Figure 11 See https://www.vishay.com/docs/73217/an608a.pdf, page 4

In the formula (19), (Q_{GD}/V_{DS}) is a linear estimation of C_{rss}, so we will use C_{rss} directly from datasheet

Let's take an example

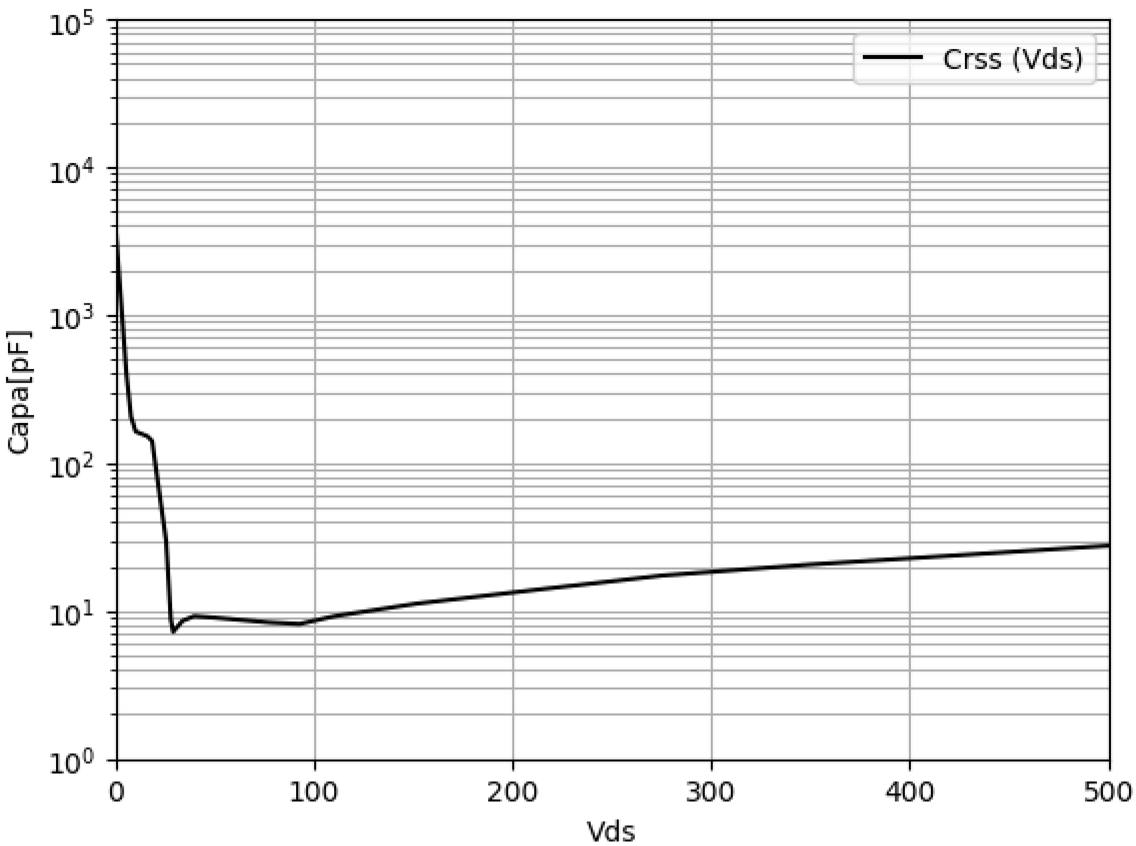
$$V_{DS} = 200 \ \ (\text{V, Example})$$
$$I_D = 10 \ \ (\text{V, Example})$$
$$R_{gExtOff} = 3 \ \ (\Omega\text{, Example})$$
$$R_{gMos} = 0.860 \ \ (\Omega\text{, see datasheet IPW60R037P7})$$
$$R_{offDriver} = 0.550 \ \ (\Omega \text{ driver datsheet ucc21520-q1})$$
$$R_{onDriver} = 5 \ \ (\Omega \text{ driver datsheet ucc21520-q1})$$
$$\text{Crss} = 13 \ \ (\text{pF @ 200V, see datasheet IPW60R037P7})$$
$$V_{gp} = 5.200 \ \ (\text{V, gate plateau, see datasheet IPW60R037P7})$$
$$V_{DS} = 200 \ \ (\text{V})$$
$$V_{TH} = 3.500 \ \ (\text{V gate threshold, see datasheet IPW60R037P7})$$
$$C_{iss} = 5000.000 \ \ (\text{pF @ 200V, see datasheet IPW60R037P7})$$

$$R_G = R_{gExtOff} + R_{offDriver} + R_{gMos} = 3 + 0.550 + 0.860 \qquad = 4.410 \ \ (\text{ohm})$$
$$V_{DG} = V_{DS} - V_{gp} = 200 - 5.200 \qquad = 194.800 \ \ (\text{V})$$

$$t_{vr} = 1 \times 10^{-3} \cdot R_G \cdot \text{Crss} \cdot \left(\frac{V_{DS}}{V_{TH}}\right) = 1 \times 10^{-3} \cdot 4.410 \cdot 13 \cdot \left(\frac{200}{3.500}\right) \qquad = 3.276 \ \ (\text{ns})$$

$$t_{ir} = 1 \times 10^{-3} \cdot R_G \cdot C_{iss} \cdot \ln\left(\frac{V_{gp}}{V_{TH}}\right) = 1 \times 10^{-3} \cdot 4.410 \cdot 5000.000 \cdot \ln\left(\frac{5.200}{3.500}\right) \qquad = 8.729 \ \ (\text{ns})$$

$$E_{off} = 1 \times 10^{-3} \cdot V_{DS} \cdot I_D \cdot \frac{t_{vr} + t_{ir}}{2} = 1 \times 10^{-3} \cdot 200 \cdot 10 \cdot \frac{3.276 + 8.729}{2} \qquad = 12.005 \ \ (\mu\text{J})$$



```
np.float64(18489.496922153317)
```

$$R_G = R_{gExtOff} + R_{offDriver} + R_{gMos} = 3 + 0.550 + 0.860 \qquad = 4.410 \ \ (\text{ohm})$$
$$V_{DG} = V_{DS} - V_{gp} = 200 - 5.200 \qquad = 194.800 \ \ (\text{V})$$
$$\text{Crss} = \frac{Q_{GD}}{V_{DS}} = \frac{36000.000}{200} \qquad = 180.000 \ \ (\text{pF})$$

$$t_{vr} = 1 \times 10^{-3} \cdot R_G \cdot \text{Crss} \cdot \left(\frac{V_{DS}}{V_{TH}}\right) = 1 \times 10^{-3} \cdot 4.410 \cdot 180.000 \cdot \left(\frac{200}{3.500}\right) \qquad = 45.360 \ \ (\text{ns})$$

$$t_{ir} = 1 \times 10^{-3} \cdot R_G \cdot C_{iss} \cdot \ln\left(\frac{V_{gp}}{V_{TH}}\right) = 1 \times 10^{-3} \cdot 4.410 \cdot 5000.000 \cdot \ln\left(\frac{5.200}{3.500}\right) \qquad = 8.729 \ \ (\text{ns})$$

$$E_{off} = 1 \times 10^{-3} \cdot V_{DS} \cdot I_D \cdot \frac{t_{vr} + t_{ir}}{2} = 1 \times 10^{-3} \cdot 200 \cdot 10 \cdot \frac{45.360 + 8.729}{2} \qquad = 54.089 \ \ (\mu\text{J})$$

for R_gExtOff in [2, 3,5,10]: R_G = R_gExtOff +R_offDriver+R_gMos # ohm a=[] idarr = range(1, 11, 1) for I_D in idarr: t_vr= R_G*1e-3*Crss*(V_DS/V_TH) # ns t_ir = R_G * 1e-3*C_iss *log(V_gp/V_TH)# ns E_off = 1e-3*V_DS*I_D*(t_vr+t_ir)/2 # µJ a.append(E_off) plt.plot(idarr, a, label = f'{R_gExtOff }') plt.legend()

$$t_{if} = R_G \cdot C_{iss} \cdot \ln\left[\left(1 + \frac{g_{fs}L_S}{R_G C_{iss}}\right) \times \frac{V_{gp}}{V_{TH}}\right]$$

$$g_{fs} = \frac{\Delta I_D}{\Delta V_{GS}}$$

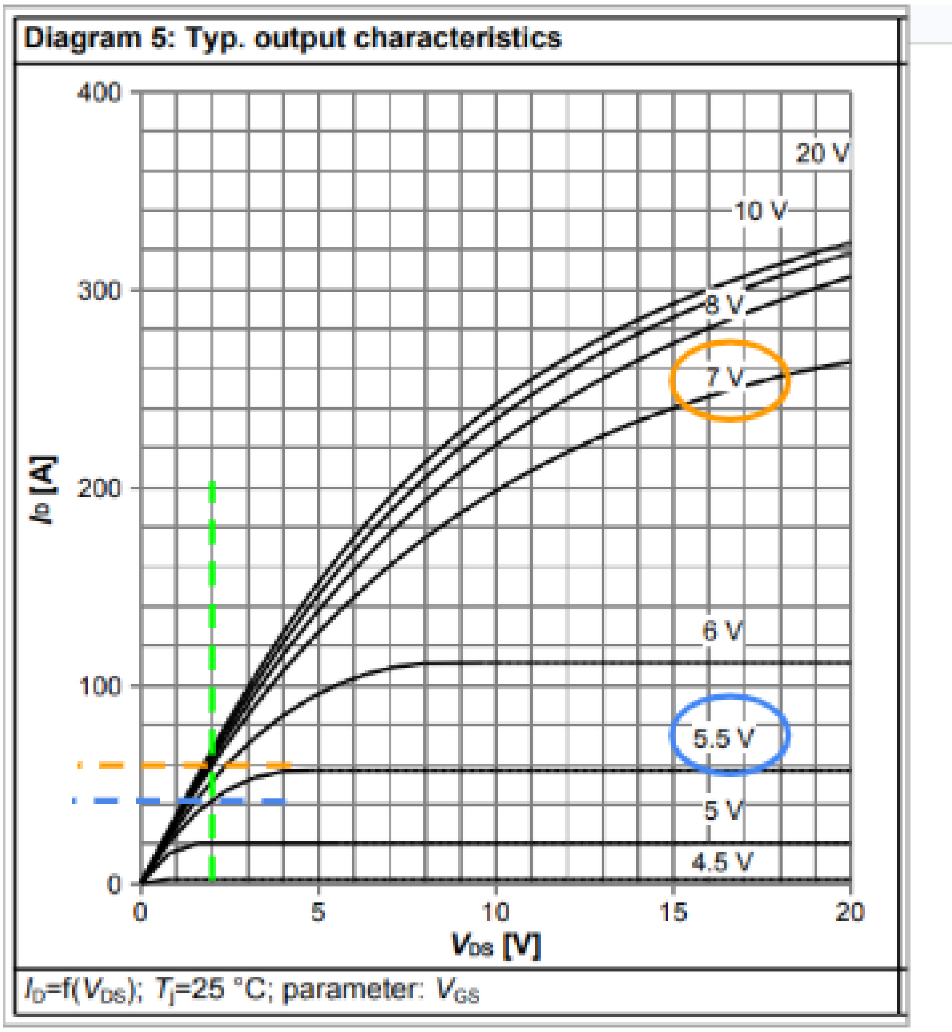


Figure 12 See https://www.vishay.com/docs/73217/an608a.pdf, page 4

$$\Delta_I = 60 - 42 = 18 \quad (\text{A @ 25°C})$$

$$\Delta_V = 7 - 5.5 = 1.500 \quad (\text{V @ 25°C})$$

$$\text{gfs} = \frac{\Delta_I}{\Delta_V} = \frac{18}{1.500} = 12.000$$

Delta_I = 98 # A Delta_V = 3 gfs= Delta_I/Delta_V R_gExtOff=3 # Ω R_G = R_gExtOff +R_offDriver+R_gMos # Ω

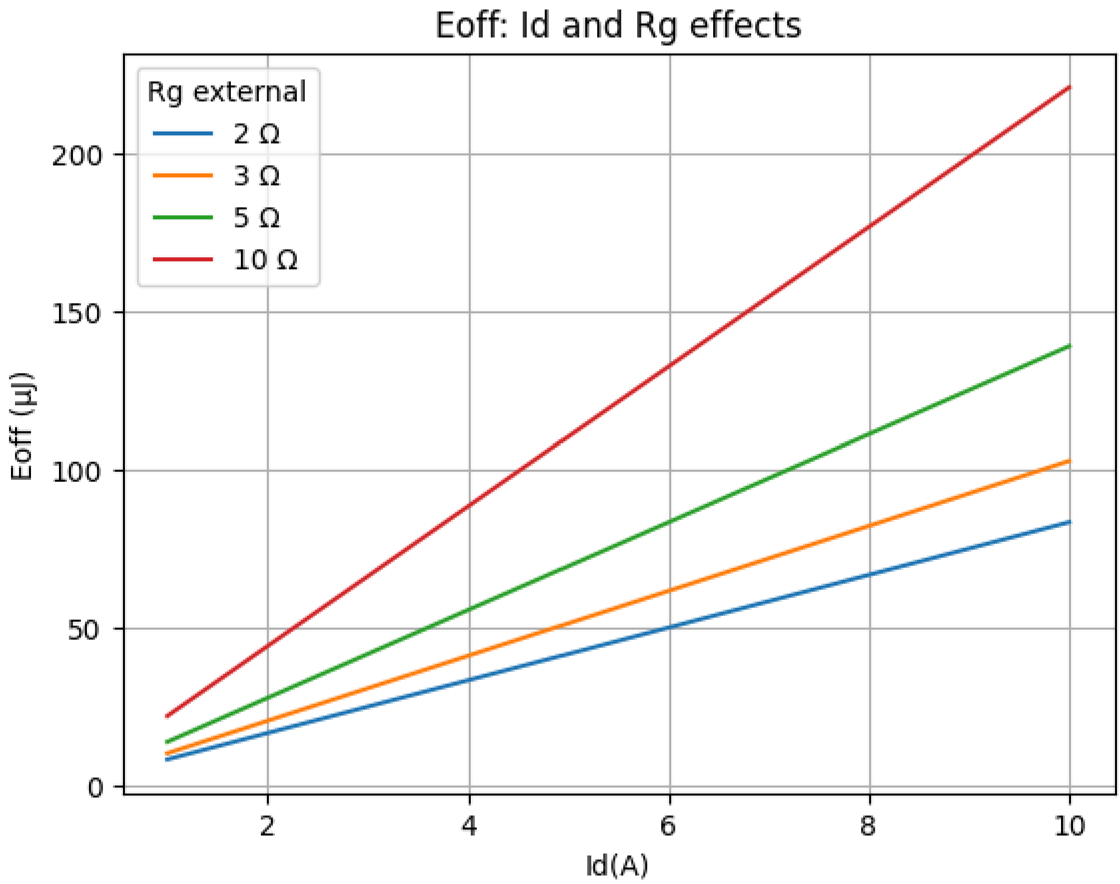
$$\text{Ls} = 15000.000 \quad (\text{pF})$$

$$t_{ir} = 1 \times 10^{-3} \cdot R_G \cdot C_{iss} \cdot \ln\left(\left(1 + \left(\text{gfs} \cdot \frac{\text{Ls}}{R_G \cdot C_{iss}}\right)\right) \cdot \frac{V_{gp}}{V_{TH}}\right)$$

$$= 1 \times 10^{-3} \cdot 4.410 \cdot 5000.000 \cdot \ln\left(\left(1 + \left(12.000 \cdot \frac{15000.000}{4.410 \cdot 5000.000}\right)\right) \cdot \frac{5.200}{3.500}\right)$$

$$= 57.575 \quad (\text{ns})$$

$$E_{off} = 1 \times 10^{-3} \cdot V_{DS} \cdot I_D \cdot \frac{t_{vr} + t_{ir}}{2} = 1 \times 10^{-3} \cdot 200 \cdot 10 \cdot \frac{45.360 + 57.575}{2} = 102.935 \quad (\mu\text{J})$$



Simulation of the double pulse test: SW-OFF

Simulation simulation of the SW-OFF

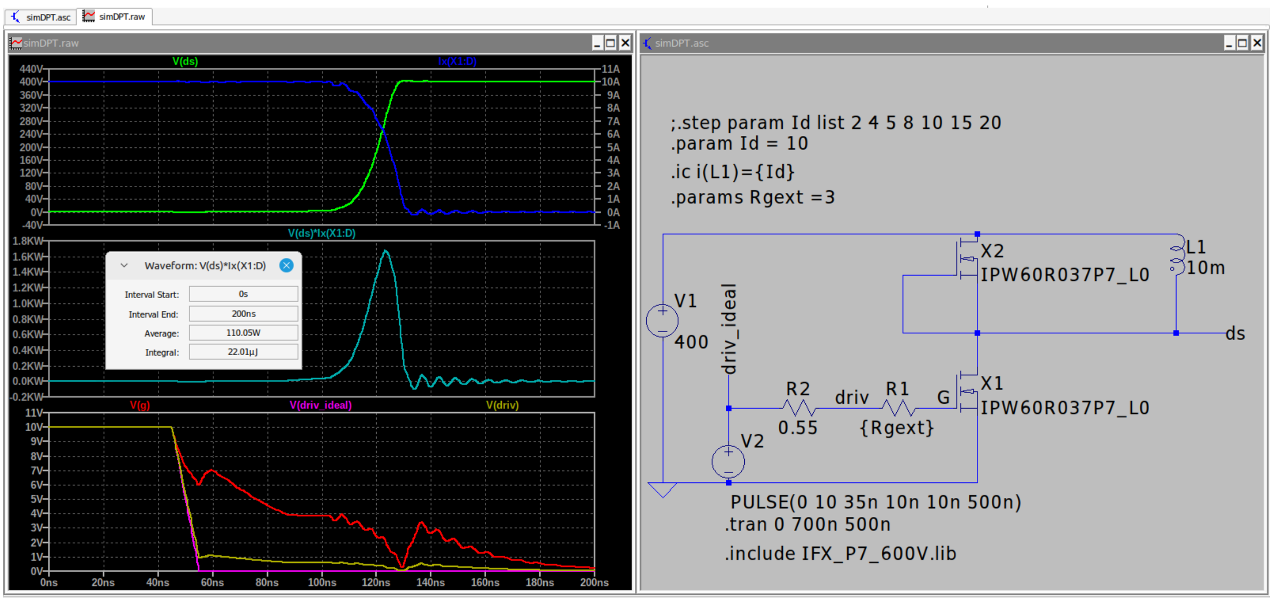


Figure 13 Simple simulation : DPT, Rg = 3 ohm, Id =10A => Eoff = 22μJ

Drain current effect

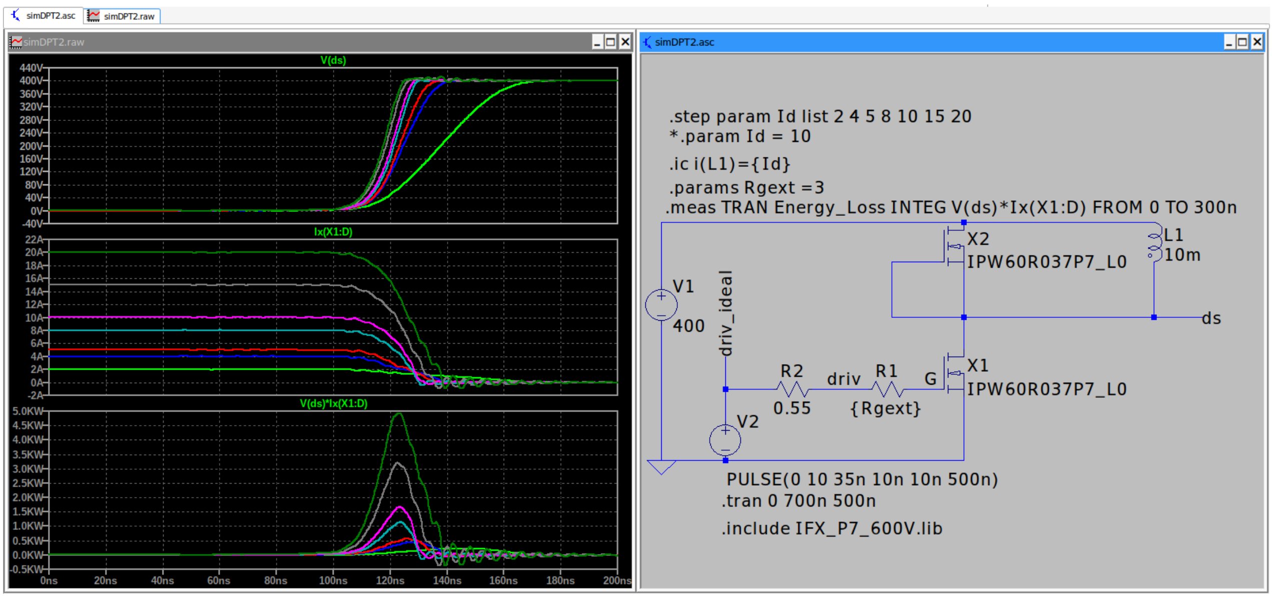
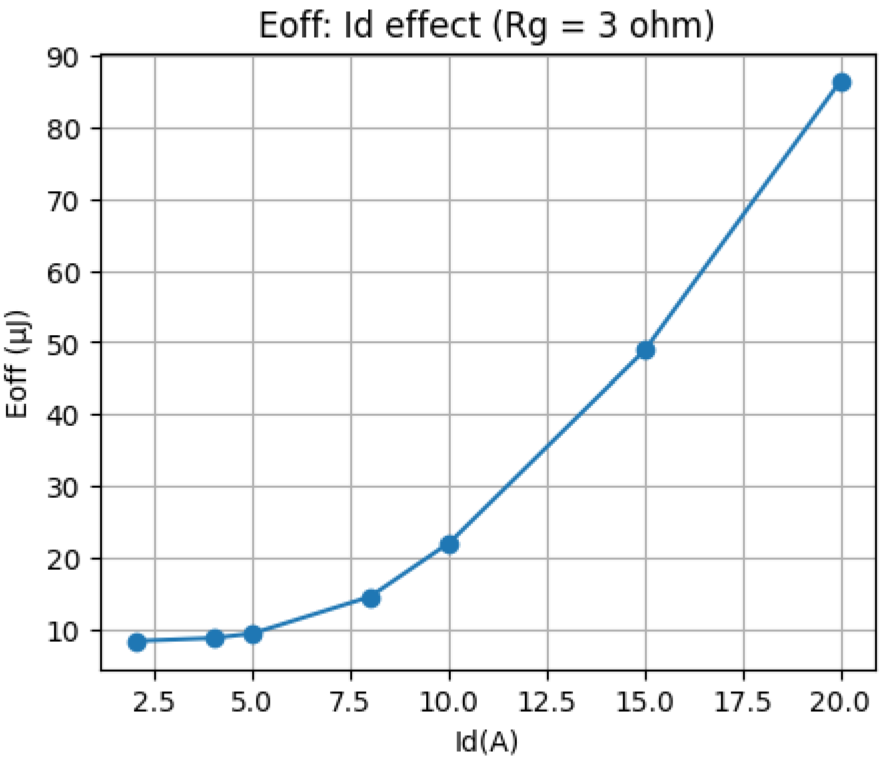


Figure 14 Simple simulation : DPT, Rg = 3 ohm, Id = [2,4,8,10,15,20A]

For more information about integral in LSTPICE step cmd see the link

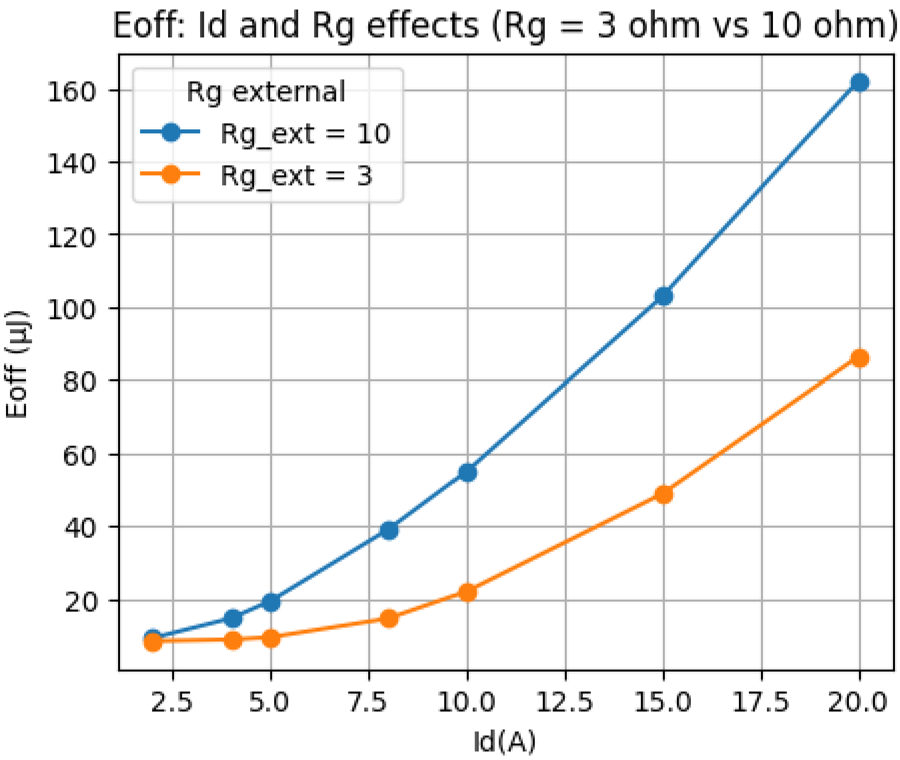
views > output log

```
.step id=2
.step id=4
.step id=5
.step id=8
.step id=10
.step id=15
.step id=20
Measurement: energy_loss
   step  INTEG(V(ds)*Ix(X1:D) )  FROM    TO
     1  8.38741496823e-06       0       3e-07
     2  8.80600287605e-06       0       3e-07
     3  9.44361803342e-06       0       3e-07
     4  1.45722776165e-05       0       3e-07
     5  2.20096978946e-05       0       3e-07
     6  4.88741641517e-05       0       3e-07
     7  8.64114082734e-05       0       3e-07
```

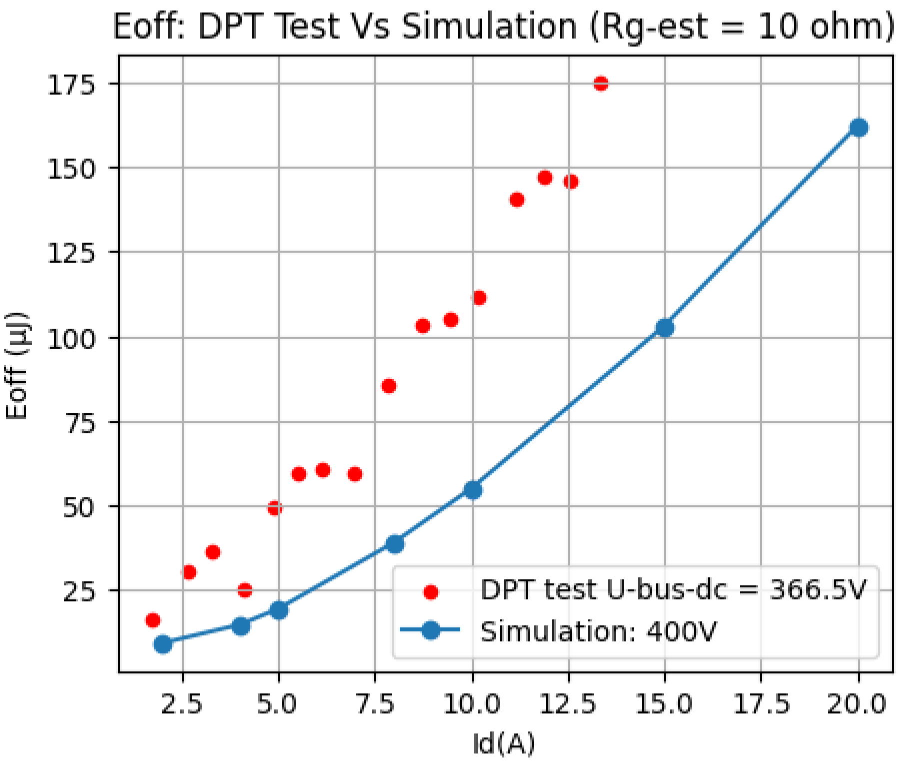


Drain current effect: compare 2 gate resistors

```
.step id=2
.step id=4
.step id=5
.step id=8
.step id=10
.step id=15
.step id=20
Measurement: energy_loss
  step  INTEG(V(ds)*Ix(X1:D) )  FROM    TO
     1  9.23653025578e-06       0       5e-07
     2  1.46592529163e-05       0       5e-07
     3  1.93705171833e-05       0       5e-07
     4  3.89830410034e-05       0       5e-07
     5  5.49416954562e-05       0       5e-07
     6  0.000102951972159       0       5e-07
     7  0.000162117103596       0       5e-07
```



Below a comparaison between the actual simulation and a double pulses test (DPT), for more information about this test and other mosfet/driver test, please chech the PDF or the folder.



We can see that the simulation model has the same signature as the test, but it **underestimate** the switching loss

Gate resistor effect

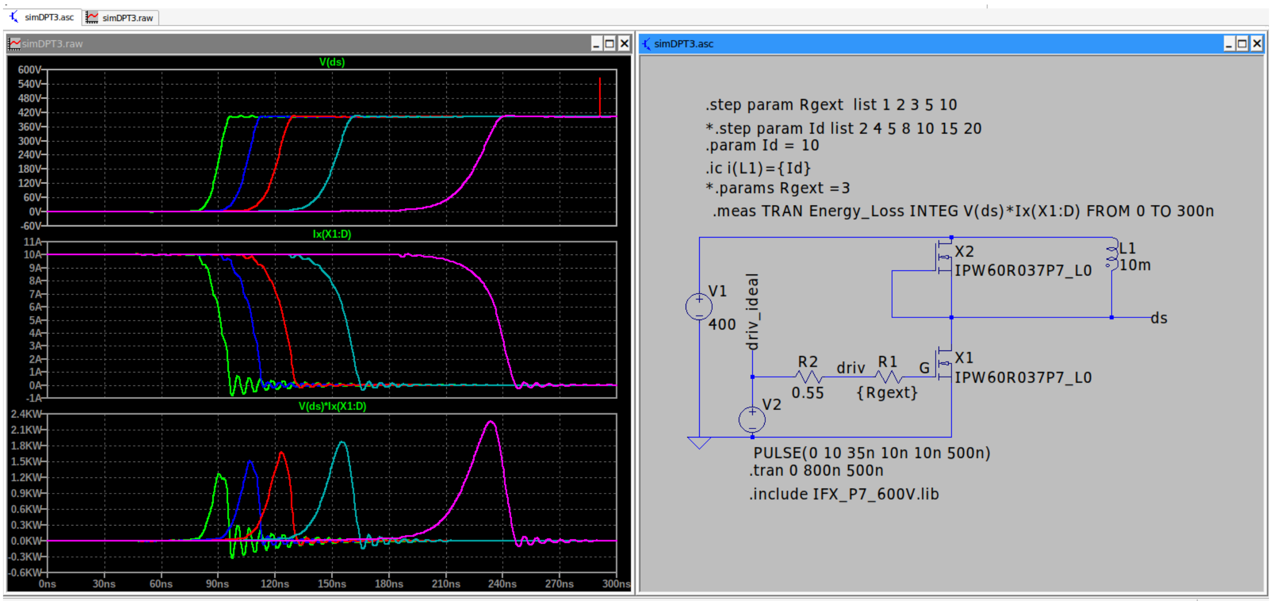
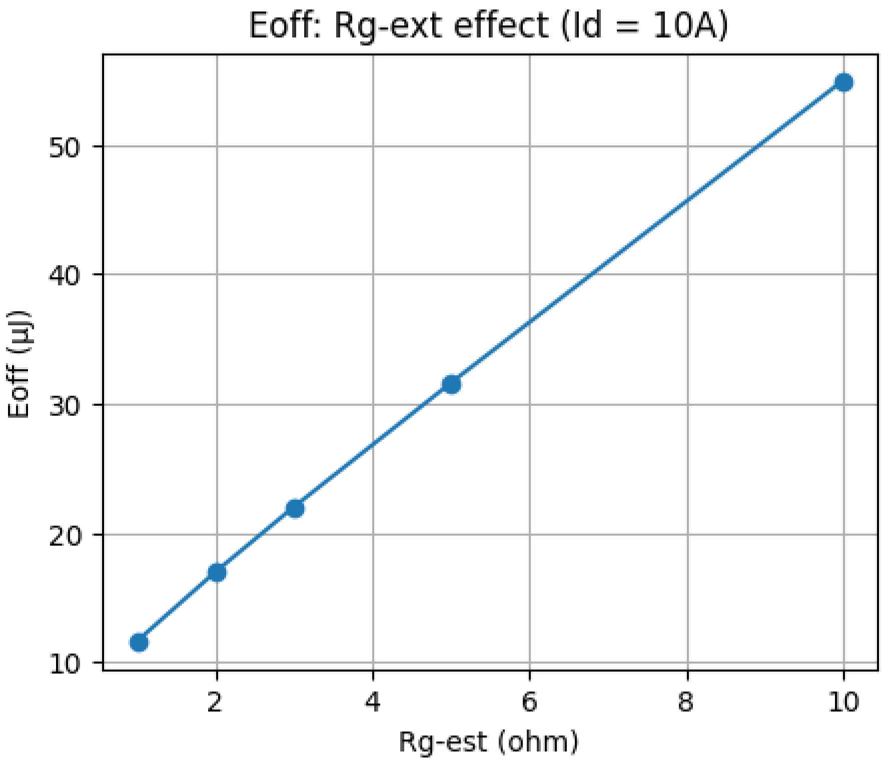


Figure 15 Simple simulation : DPT, Rg = [1,2,3,5,10] ohm, Id = 10A

```
.step rgext=1
.step rgext=2
.step rgext=3
.step rgext=5
.step rgext=10
Measurement: energy_loss
  step  INTEG(V(ds)*Ix(X1:D) )  FROM    TO
     1  1.16431044375e-05       0       3e-07
     2  1.69911850135e-05       0       3e-07
     3  2.20091411318e-05       0       3e-07
     4  3.15740994641e-05       0       3e-07
     5  5.49211220357e-05       0       3e-07
```



Low gate resistor make switch off fast and reduce the loss.

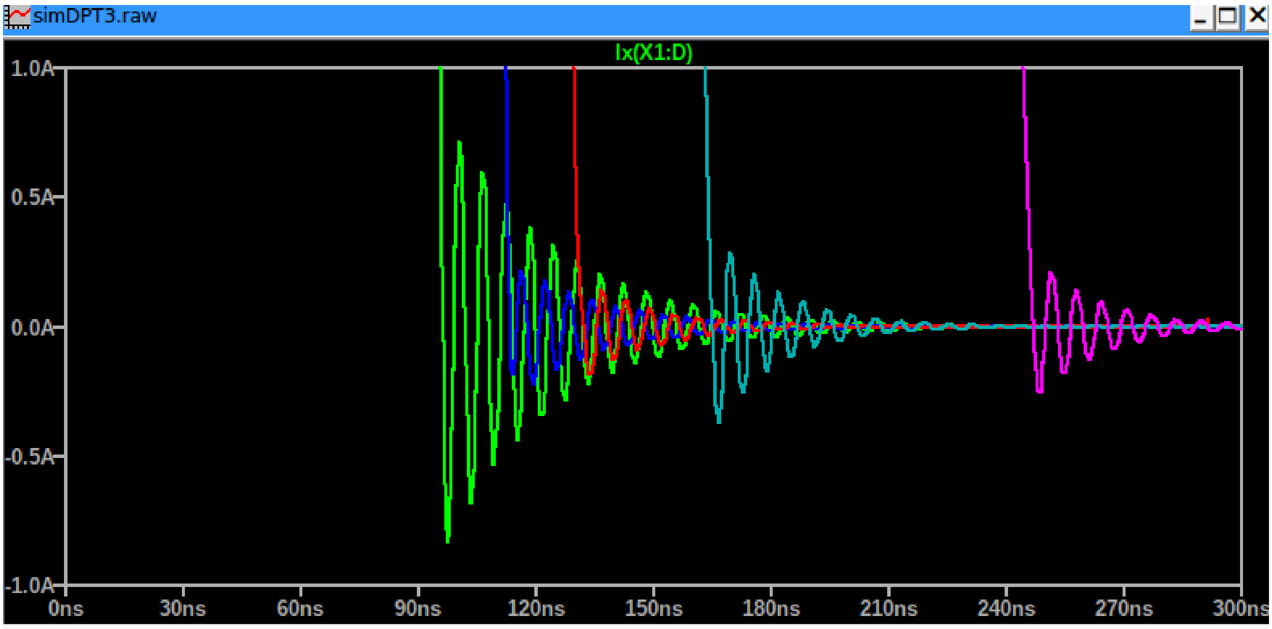


Figure 16 Drain current oscillation, Simple simulation : DPT, Rg = [1,2,3,5,10] ohm, Id = 10A

We can see in the above zoom that low gate resistor (green) make switching fast, but increase Id oscillation (same for Vds).

So, an optimal value for R_gate must therefore be determined: lying between a low value, which reduces switching losses but causes significant oscillations, and a high value, which increases losses but dampens those oscillations.

Gate loop inductance effect

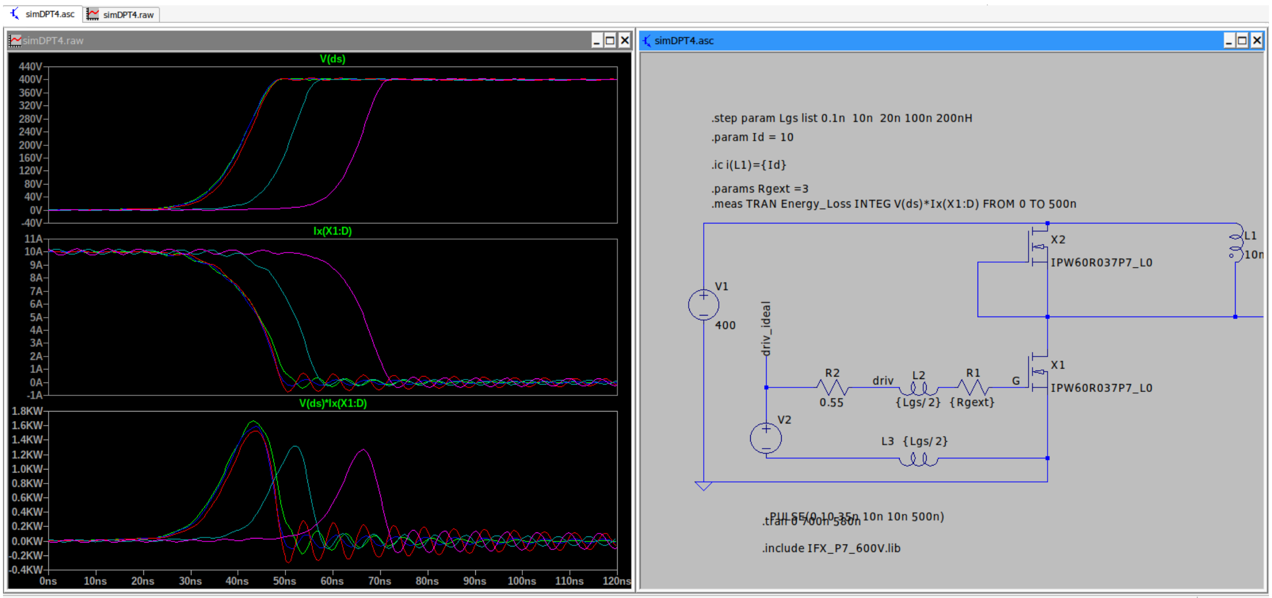
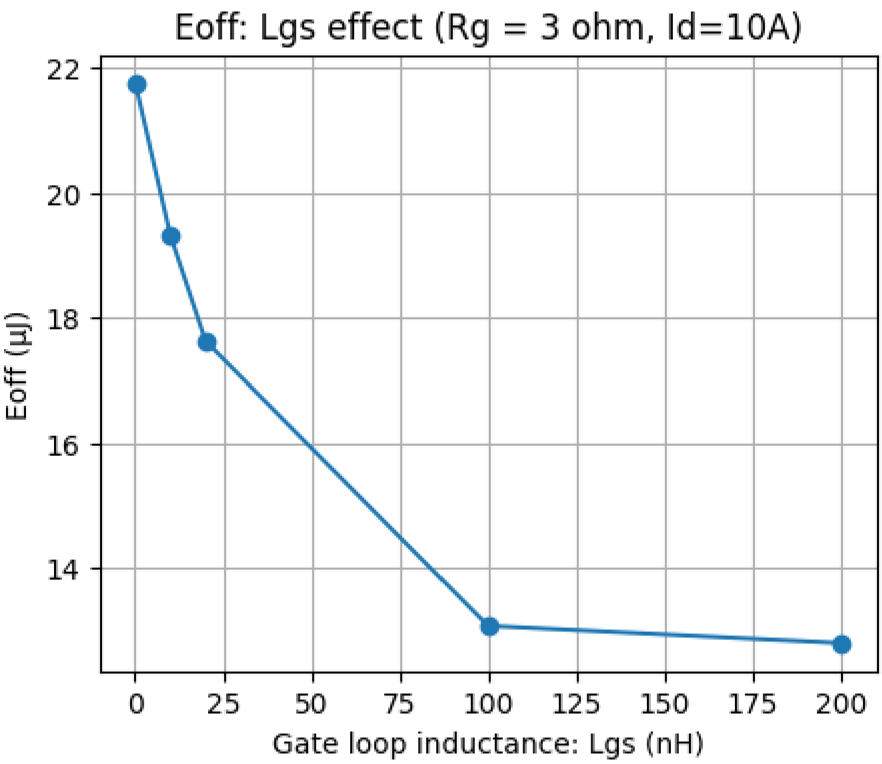


Figure 17 Gate inductance effect

```
.step lgs=1e-10
.step lgs=1e-08
.step lgs=2e-08
.step lgs=1e-07
.step lgs=2e-07
Measurement: energy_loss
  step  INTEG(V(ds)*Ix(X1:D) )  FROM    TO
     1  2.17564101527e-05       0       5e-07
     2  1.93255164435e-05       0       5e-07
     3  1.76290608097e-05       0       5e-07
     4  1.3076171685e-05        0       5e-07
     5  1.28007732693e-05       0       5e-07
```



For margin we will take Eoff = 120µJ as the maximum for the next of this study:

$$\mathrm{Eoff} = 120 \ \ (\mu\mathrm{J})$$
$$f_{sw} = 150 \ \ (\mathrm{kHz}\ ,\ \mathrm{max})$$
$$P_{sw} = 1 \times 10^{-3} \cdot \mathrm{Eoff} \cdot f_{sw} = 1 \times 10^{-3} \cdot 120 \cdot 150 \qquad = 18.000 \ \ (\mathrm{W})$$

Mosfet loss (conduction and swiching)

Reminder : IPW60R037P7 600V 76A 37mΩ

for mor information, see the Mosfet datasheet.

Drain-source on-state resistance RDS(on)

- 0.030Ω : VGS=10V,ID=29.5A,Tj=25°C
- 0.070Ω : VGS=10V,ID=29.5A,Tj=150°C

From the initial calculation of the LLC thank , we have the rms in primary is:

$$\mathrm{Ir}_{rms} = 10.354 \ \ (\mathrm{Arms})$$

So the rms current in the moset is

$$I_{Q\_rms} = \sqrt{\frac{1}{2} I_{r\_rms}^2} = \frac{I_{r\_rms}}{\sqrt{2}}$$

Calculation

$$I_{Qrms} = \frac{\mathrm{Ir}_{rms}}{\sqrt{2}} = \frac{10.354}{\sqrt{2}} \qquad = 7.321 \ \ (\mathrm{Arms})$$
$$I_{Qpeak} = I_{Qrms} \cdot \sqrt{2} = 7.321 \cdot \sqrt{2} \qquad = 10.354 \ \ (\mathrm{Arms})$$

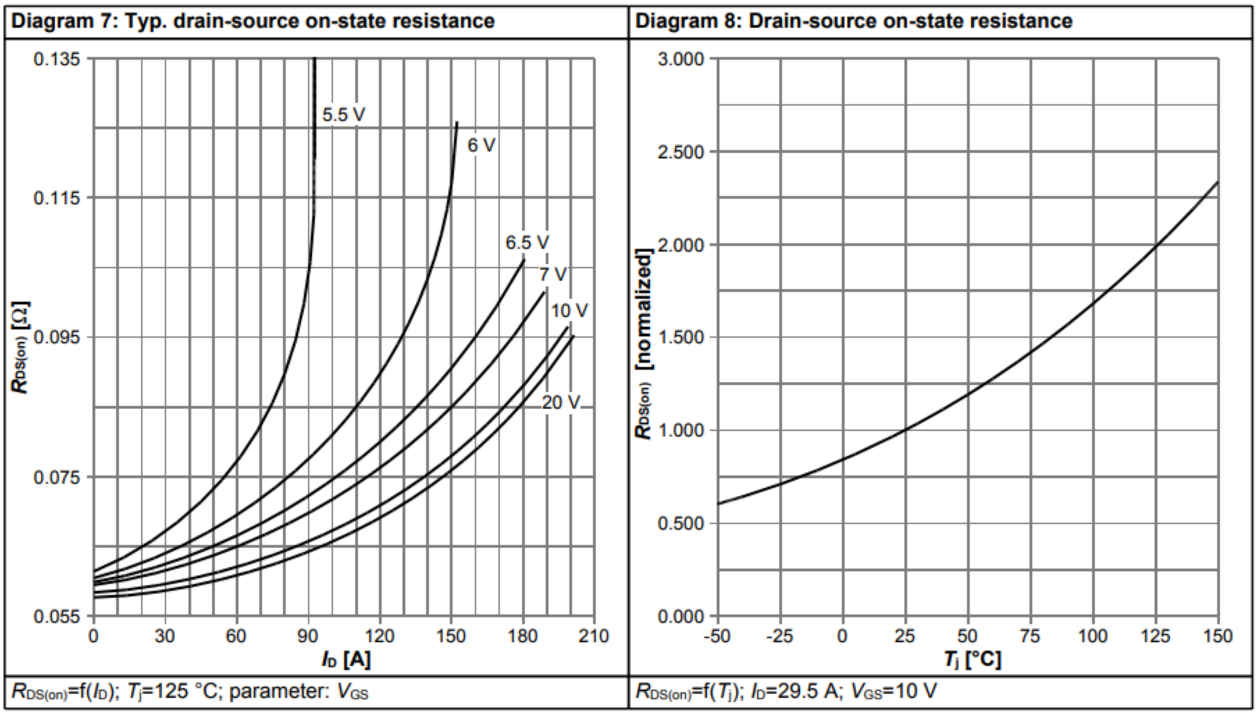


Figure 18 Rds-on of IPW60R037P7

$$R_{dson} = 60 \;\; (\text{m}\Omega \text{ @Tj} = 125°\text{C and Id}= 11\text{A and vds} =10\text{V})$$

The conduction loss by mos

$$P_{cond} = 1 \times 10^{-3} \cdot R_{dson} \cdot \left(I_{Qrms}\right)^2 = 1 \times 10^{-3} \cdot 60 \cdot (7.321)^2 \qquad = 3.216 \;\; (\text{W})$$

Total loss by mosfet

$$P_{mos} = P_{sw} + P_{cond} = 18.000 + 3.216 \qquad = 21.216 \;\; (\text{W})$$

MOSFET Datasheet (Rth):

- Thermal resistance, junction - case Rth_JC : 0.49 °C/W -
- Thermal resistance, junction - ambient Rth_JA: 62 °C/W

Without heatsink

$$\text{Rth}_{JA} = 62 \;\; (°\text{C/W})$$
$$\Delta_{Tja} = \text{Rth}_{JA} \cdot P_{mos} = 62 \cdot 21.216 \qquad = 1315.402 \;\; (°\text{C})$$

So, using mosfet without heatsink is not possible

With heatsink

$$\text{Rth}_{JC} = 0.490 \;\; (°\text{C/W})$$
$$\Delta_{Tjc} = \text{Rth}_{JC} \cdot P_{mos} = 0.490 \cdot 21.216 \qquad = 10.396 \;\; (°\text{C})$$

The delta T is manageable, but we must check the path of heat flux between the case until the air

Figure 19 Sil thermal used to couple the mosfets to the heatsink

- Thickness = 300µm (4 x 0.3mm = 1.2mm)
- Thermal conductivity = 1.2 W/°C

$$\begin{aligned}
\mathrm{Rth}_{JC} &= 0.490 \ \ (°\mathrm{C/W}) \\
\mathrm{Rth}_{SilTherm} &= 1.200 \ \ (°\mathrm{C/W}) \\
\mathrm{Rth}_{CS} &= \mathrm{Rth}_{JC} + \mathrm{Rth}_{SilTherm} = 0.490 + 1.200 \qquad = 1.690 \ \ (°\mathrm{C/W\ joncton\ to\ sink}) \\
\Delta_{Tjs} &= \mathrm{Rth}_{CS} \cdot P_{mos} = 1.690 \cdot 21.216 \qquad = 35.855 \ \ (°\mathrm{C})
\end{aligned}$$

Quick reminder of the maximum dissipated power

In the datasheet we found this information:

- Power dissipation Ptot 255 W @ TC=25°C

This is a theorical maximum dissipation total power, it is calculated using the below formula:

$$\begin{aligned}
\mathrm{Rth}_{JC} &= 0.490 \ \ (°\mathrm{C/W,\ datasheet}) \\
T_{case} &= 25 \ \ (°\mathrm{C,\ datasheet}) \\
T_{jmax} &= 150 \ \ (°\mathrm{C,\ datasheet}) \\
P_{dissMax} &= \frac{T_{jmax} - T_{case}}{\mathrm{Rth}_{JC}} = \frac{150 - 25}{0.490} \qquad = 255.102 \ \ (\mathrm{W})
\end{aligned}$$

There is therefore a direct link to the junction-to-case thermal resistance, given that the maximum junction temperature of almost all MOSFETs are in the range of 150 - 175°C. So a mosfet with heigh "maximum power dissipation" mean lower junction-to-case thermal resistance, so easy to cold

We can use the same formula to calculate the maximum continue current of the mosfet

At case temperature of 25°C

$$
\begin{aligned}
T_{case} &= 25 \ \ (°C) \\
T_j &= 150 \ \ (°C) \\
P_{dissMax} &= \frac{T_j - T_{case}}{\mathrm{Rth}_{JC}} = \frac{150 - 25}{0.490} && = 255.102 \ \ (W) \\
\mathrm{Id}_{100} &= \sqrt{\frac{P_{dissMax}}{0.037}} = \sqrt{\frac{255.102}{0.037}} && = 83.034 \ \ (\text{A @ Tj= 100°C, datasheet 76 A})
\end{aligned}
$$

At case temperature of 100°C

$$
\begin{aligned}
T_{case} &= 100 \ \ (°C) \\
T_j &= 150 \ \ (°C) \\
P_{dissMax} &= \frac{T_j - T_{case}}{\mathrm{Rth}_{JC}} = \frac{150 - 100}{0.490} && = 102.041 \ \ (W) \\
\mathrm{Id}_{100} &= \sqrt{\frac{P_{dissMax}}{0.037}} = \sqrt{\frac{102.041}{0.037}} && = 52.515 \ \ (\text{A @ Tj= 100°C, datasheet 48 A})
\end{aligned}
$$

The calculations above overestimate the maximum continuous drain current because we used a constant value for Rds; in reality, Rds varies with junction temperature (Tj). However, our estimate remains a good first-order approximation of this current.

$$
\begin{aligned}
\mathrm{Id}_{Datasheet25} &= 76 \ \ (A) \\
\mathrm{Id}_{Estimation25} &= 83 \ \ (A) \\
\mathrm{Error}_{25} &= 100 \cdot \frac{\mathrm{Id}_{Estimation25} - \mathrm{Id}_{Datasheet25}}{\mathrm{Id}_{Datasheet25}} = 100 \cdot \frac{83 - 76}{76} && = 9.211 \ \ (\backslash\%) \\
\mathrm{Id}_{Datasheet100} &= 48 \ \ (A) \\
\mathrm{Id}_{Estimation100} &= 52.500 \ \ (A) \\
\mathrm{Error}_{100} &= 100 \cdot \frac{\mathrm{Id}_{Estimation100} - \mathrm{Id}_{Datasheet100}}{\mathrm{Id}_{Datasheet100}} = 100 \cdot \frac{52.500 - 48}{48} && = 9.375 \ \ (\backslash\%)
\end{aligned}
$$

Our overstimation is around 9.3%

Gate driver

https://assets.nexperia.com/documents/application-note/AN90059.pdf

https://www.ti.com/lit/ds/symlink/ucc21520-q1.pdf

Gate driver power consumption

$$E_G = \int_{t_0}^{T} V_{GS}(t)\, I_G(t)\, dt = V_{GS} \int_{t_0}^{T} I_G(t)\, dt = V_{GS} \cdot Q_{gTot}$$

$$Q_{gTot} = 127 \ \ (\text{nC})$$
$$V_{GS} = 12 \ \ (\text{V})$$
$$F_{sw} = 150 \ \ (\text{kHz})$$
$$E_G = V_{GS} \cdot Q_{gTot} = 12 \cdot 127 \qquad = 1524 \ \ (\text{nJ})$$
$$P_G = F_{sw} \cdot E_G \cdot 1 \times 10^{-3} = 150 \cdot 1524 \cdot 1 \times 10^{-3} \qquad = 228.600 \ \ (\text{mW})$$
$$P_{gLeg} = 2 \cdot P_G = 2 \cdot 228.600 \qquad = 457.200 \ \ (\text{mW})$$

Let's calculat the power dissipation of the driver (ON/OFF/Total)

$$R_{gMos} = 0.860 \ \ (\Omega \text{ Mosfet datasheet})$$
$$R_{onDriver} = 5 \ \ (\Omega \text{ driver datsheet})$$
$$R_{offDriver} = 0.550 \ \ (\Omega \text{ driver datsheet})$$
$$R_{gExr} = 3 \ \ (\Omega \text{ example})$$

$$\begin{aligned} P_{dissDriverOn} &= P_{gLeg} \cdot \frac{R_{onDriver}}{R_{onDriver} + R_{gExr} + R_{gMos}} \\ &= 457.200 \cdot \frac{5}{5 + 3 + 0.860} \\ &= 258.014 \ \ (\text{mW}) \end{aligned}$$

$$\begin{aligned} P_{dissDriverOff} &= P_{gLeg} \cdot \frac{R_{offDriver}}{R_{onDriver} + R_{gExr} + R_{gMos}} \\ &= 457.200 \cdot \frac{0.550}{5 + 3 + 0.860} \\ &= 28.381 \ \ (\text{mW}) \end{aligned}$$

$$\begin{aligned} P_{dissDriver} &= P_{dissDriverOn} + P_{dissDriverOff} \\ &= 258.014 + 28.381 \\ &= 286.395 \ \ (\text{mW}) \end{aligned}$$

OK , since we have the marging with the driver maximum power dissipation (both sides) 950 mW

Using the junction-to-ambient thermal resistance :

$$R_{thJA} = 69.800 \ \ (°\text{C/W})$$
$$\Delta_T = 1 \times 10^{-3} \cdot P_{dissDriver} \cdot R_{thJA} = 1 \times 10^{-3} \cdot 286.395 \cdot 69.800 \qquad = 19.990 \ \ (°\text{C})$$

The same calculation of other external gate resistor values:

print(df.to_string(index=False, formatters={c: "{:.0f}".format for c in df.columns}))

So driver is safe even with 1 Ω of external gate resistor

Bootstrap circuit

Boostrap capacitor

$$\Delta_{Vdd} = 0.500 \ \ (\mathrm{V})$$
$$Q_{gTot} = 127 \ \ (\mathrm{nC})$$
$$C_{boot} = \frac{Q_{gTot}}{\Delta_{Vdd}} = \frac{127}{0.500} \qquad = 254.000 \ \ (\mathrm{nC})$$

Since there is other charge consummators like Qrr of the bootstrap diode, level shift charge, and other current, we will take a value of

$$C_{bootMarg} = C_{boot} \cdot 4 = 254.000 \cdot 4 \qquad = 1016.000 \ \ (\mathrm{uF})$$

We will shose 1.1µF = 1µF // 100nF (Conservative design rules recommend factor of x10 to x20)

Boostrap resistor

$$R_{boot} = 2 \ \ (\Omega)$$
$$V_{dd} = 12 \ \ (\mathrm{V})$$
$$I_{bootPeak} = 8 \ \ (\mathrm{A})$$
$$R_{bootMin} = \frac{V_{dd}}{I_{bootPeak}} = \frac{12}{8} \qquad = 1.500 \ \ (\Omega)$$

Let's take $R_{boot} = 2.15\Omega$

To fully charge the capacitor to 99% of V_{DD}, you need at least 5τ of time.

$$C_{boot} = 1.100 \ \ (\mu\mathrm{F})$$
$$R_{boot} = 2.150 \ \ (\Omega)$$
$$\tau = C_{boot} \cdot R_{boot} = 1.100 \cdot 2.150 \qquad = 2.365 \ \ (\mu\mathrm{s})$$
$$T_{Charge} = 5 \cdot \tau = 5 \cdot 2.365 \qquad = 11.825 \ \ (\mu\mathrm{s})$$

Boostrap resistance power : worst case (0.5 mean that bootsrap work only during half of periode)

$$P_{Rboot} = 0.5 \cdot 1 \times 10^{3} \cdot \frac{\left(\Delta_{Vdd}\right)^{2}}{R_{boot}} = 0.5 \cdot 1 \times 10^{3} \cdot \frac{(0.500)^{2}}{2.150} \qquad = 58.140 \ \ (\mathrm{mW})$$

Why this is the worst case:

- This formula suppose that the diode is perfect
- The bootstrap capa keep alwase a voltage that eqaual to $V_{dd} - \Delta_{Vdd}$

Boostrap diode

We will use US2MA diode

- DC Blocking Voltage 1000V
- Average Forward Rectified Current 1.5A

- Maximum Reverse Recovery Time = 75ns
- Typical Junction Capacitance 30pF
- Peak Forward Surge Current (8.3ms) 50A

for more information see the datahsheet

Secondary

Mosfet

IRFP4568PBF 150 V 171A 4.8mΩ

IRFP4568PBF Datasheet

MOSFET power losses

Since the secondary is synchron rectifier (SRC), so the mosfet achieve natural ZVS (zero current switching), so switching losses is almost 0 W.

From the initial calculation of the LLC thank , we have the the input data for primary mosfet calculation:

```
Lnc              3.0
Qec              0.55
Cr_nF            116.209
n                4.0
Lr_uH            21.797
Lm_uH            65.392
fsw_min          60170.0
fsw_max          156220.0
Im_rms           6.992
Io               25.0
Ioe_rms          7.636
Ios_rms          30.545
Ir_rms           10.354
L_second_uH      4.087
Re_nom           24.901
Re_110           22.637
Cr               1.16e-07
Lr               2.18e-05
Lm               6.54e-05
```

We have the rms current in secondary side:

$$\mathrm{Ir}_{rms} = 30.545 \ \ \mathrm{(Arms)}$$

So the rms current in the moset is

$$I_{Q\_rms} = \sqrt{\frac{1}{2} I^2_{r\_rms}} = \frac{I_{r\_rms}}{\sqrt{2}}$$

Calculation

$$I_{Qrms} = \frac{\mathrm{Ir}_{rms}}{\sqrt{2}} = \frac{30.545}{\sqrt{2}} \qquad = 21.599 \;\; (\mathrm{Arms})$$

$$I_{Qpeak} = I_{Qrms} \cdot \sqrt{2} = 21.599 \cdot \sqrt{2} \qquad = 30.545 \;\; (\mathrm{Arms})$$

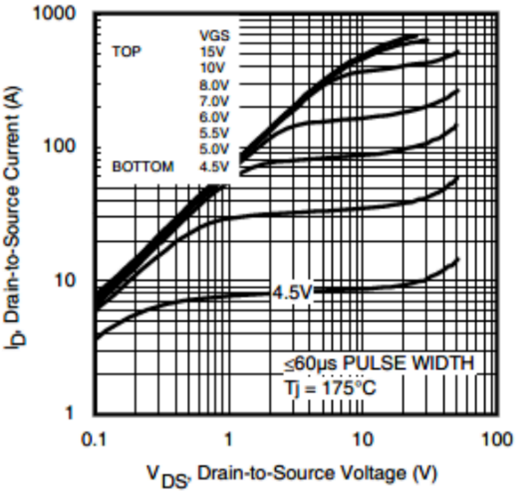


Figure 20

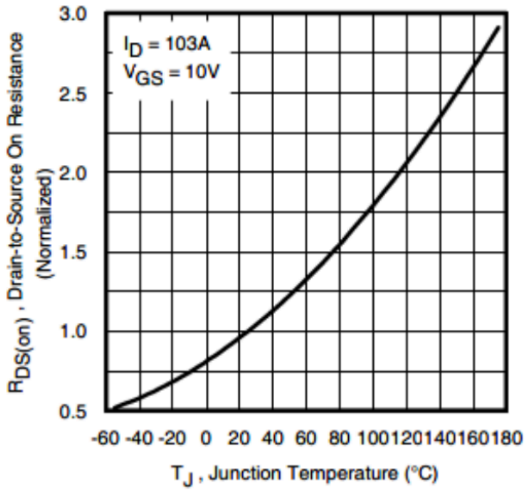


Figure 21

At [Id,Vgs] = [30A, 10V] and Tj = 175°C we have Vds = 0.4V(above, left figure)

$$\mathrm{Id} = 30 \;\; (\mathrm{A})$$

$$\mathrm{Vds} = 0.400 \;\; (\mathrm{V})$$

$$\mathrm{Rds}_{175} = \frac{\mathrm{Vds}}{\mathrm{Id}} = \frac{0.400}{30} \qquad = 0.013 \;\; (\Omega)$$

let's use the above right figure to calculate Rds @ 75°C

$$\mathrm{Rds}_{Norm175} = 2.800 \;\; (@\ 175°\mathrm{C})$$

$$\mathrm{Rds}_{Norm75} = 1.500 \;\; (@\ 75°\mathrm{C})$$

$$\mathrm{Rds}_{75} = \mathrm{Rds}_{175} \cdot \left(\frac{\mathrm{Rds}_{Norm75}}{\mathrm{Rds}_{Norm175}}\right) = 0.013 \cdot \left(\frac{1.500}{2.800}\right) \qquad = 0.007 \;\; (\Omega)$$

Conduction loss in each mosfet @ 75°C

$$P_{cond} = \left(I_{Qrms}\right)^2 \cdot \mathrm{Rds}_{75} = (21.599)^2 \cdot 0.007 \qquad = 3.332 \;\; (\mathrm{W})$$

From the mosfet datasheet

- Rth_JC Junction-to-Case 0.29 °C/W
- Rth_CS Case-to-Sink, Flat Greased Surface 0.24 °C/W
- Rth_JA Junction-to-Ambient 40 °C/W

Without heatsink

$$Rth_{JA} = 40 \ \ (°C/W)$$
$$\Delta_{Tja} = Rth_{JA} \cdot P_{cond} = 40 \cdot 3.332 \qquad = 133.285 \ \ (°C)$$

So, using mosfet without heatsink is hard (133°C of delta is a lot)

With heatsink

$$Rth_{JC} = 0.290 \ \ (°C/W)$$
$$\Delta_{Tjc} = Rth_{JC} \cdot P_{cond} = 0.290 \cdot 3.332 \qquad = 0.966 \ \ (°C)$$

The delta T is manageable, but we must check the path of heat flux between the case until the air

Figure 22 Sil thermal used to couple the mosfets to the heatsink

- Thickness = 300µm (4 x 0.3mm = 1.2mm)
- Thermal conductivity = 1.2 W/°C

$$Rth_{JC} = 0.290 \ \ (°C/W)$$
$$Rth_{SilTherm} = 1.200 \ \ (°C/W)$$
$$Rth_{CS} = Rth_{JC} + Rth_{SilTherm} = 0.290 + 1.200 \qquad = 1.490 \ \ (°C/W \text{ joncton to sink})$$
$$\Delta_{Tjs} = Rth_{CS} \cdot P_{cond} = 1.490 \cdot 3.332 \qquad = 4.965 \ \ (°C)$$

Driver

We will use the synchronous rectifier controller "NCP43080DDR2G" to control the secondary mosfets, for mor information, check NCP43080DDR2G datasheet.

Since the mosfet is controled only when to conduction is done by the body diode, so the on energy is close to 0, and the oscillation is very low, so the gate resistor can be 0

As the primary, we will use a selection jumper to chose between Rg of 0Ω, 1.1Ω or 3Ω